

**SPECIFICATION FOR FIRE
PROTECTION AND
PREVENTION DESIGN**

Project No.: 01000
KUS Doc No.: 01000-AC-SPC-0002
Contr. Doc No.:
Rev.: R00
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1 SCOPE

1.1 General

This specification outlines the basic principles of the building fire protection for the Karabatan Power Plant project of Karabatan Utility Solutions (KUS) located in KARABATAN, Atyrau region, Kazakhstan.

1.2 Design standard

The design and construction of the building fire protection shall be in accordance with National and Local regulations including the following:

- SNiP RK 2.02.05-2009 Fire Safety of Building and Structures
- SNiP RK 3.02.04-2009 Administrative and Welfare Buildings
- SNiP RK 3.02.09-2010 Production Buildings;
- SNiP 2.11.01-85 Storage Buildings;
- International Building code (IBC);
- International Fire code (IFC);
- National fire protection association (NFPA).

2 BUILDING FIRE PROTECTION

2.1 Fire-resistant rating

Fire-resistance ratings of the building members:

Building fire-resistance class	Fire-resistance limit (Hours)							
	Wall			Column	Escap e stair	Floor slab	Roof	
	External wall , wall of staircase and elevator	Internal wall	Room partition				Roof panel	Bearin g eleme
1	2.0	1.0	0.5	2.5	2.0	1.0	0.5	0.5

Fire-resistance ratings of the building elements:

Member	Class	Fire-resistance ratings (Hours)	Remarks
Fire wall	1	2.5	
Fire dividing wall	1	1.0	
Roller shutter	1	1.0	

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The minimum fire-ratings of the building special elements shall be as follows:

Building elements	Minimum fire rating (hours)		
	External and dividing walls ,doors and floors between buildings, warehouse, process and plant area	Internal walls and doors adjacent to escape route	Internal walls and doors between offices
Office building, including mess room, offices in plant building	2.0	1.0	0.5
Computer room	2.0	1.0	0.5
Switchgear building	2.0	2.0	0.5
Analyzer building	2.0	2.0	0.5
Heater /boiler/mechanical room	2.0	2.0	0.5
Warehouse	2.0	2.0	0.5
Control room/laboratory	2.0	1.0	1.0
Wall of stairs and elevator shaft	2.0	2.0	---

Fire-resistance rating

Fire-resistance ratings for building should be class 1.

Fire walls having at least the minimum fire resistance given below shall be installed to provide the following:

- Fire compartments separating different facilities when they are located within the same building, e.g. control room, laboratory room, switch room, etc ;
- Allow an easy evacuation of all occupants ;
- Limit the spread of the fire within and outside the building. Facilitate the access for emergency services.

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Fire walls shall be void free and shall be full height. Where opening or holes are provided for A.C. ducting, cable trays, pipes, etc., they shall be sealed, fire safe, using materials having the same fire-rating as the wall.

Materials used for walls and ceiling shall have resistance to surface spread of flame in accordance with national technical regulations. Limit the spread of the fire within and outside the building.

Fire-escape doors shall be so located that they are never more than 35 m away from any point on the floor.

The distance between the bottom of a staircase and the fire exit shall not exceed 35m.

Corridors shall not include dead-ends greater than 10m long.

Staircases shall be separated from rest of building by means of fire-resistant walls.

Interlinked fire detectors shall be placed throughout the building, in corridors meeting rooms, archives and rest rooms at a maximum centre to centre radius of 30m.

Materials used for walls and ceiling shall have resistance to surface spread of flame in accordance with national technical regulations for buildings or Euronorms whichever is the most stringent.

The materials comprising the floor walls and roof of a building shall have the minimum level of fire resistance as given herein.

2.2 The fire compartment area for production building

Fire-resistance ratings, Maximum height and occupied area of factory building:

Production class	Maximum height (m)	Fire-resistance rating	Maximum floor area of fire division(m ²)		
			Single story	multistory	
				Two stories	Three stories and up
A / B	36	1	No limit	5200	3500
A	36	2	No limit	5200	3500
B	36	2	No limit	10400	7800
C	48	1/2	No limit	25000	10400
D	54	1/2	No limit	No limit	No limit
E	54	1/2	No limit	No limit	No limit

The maximum admitted area of a fire division may be increased by one time, which an automatic sprinkler be installed.

The fire compartment wall of production building should be 2.5 hr.

The floor material of production building should be non-combustible.

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Class 1 fire-resistance rating production building, the diving wall should be 1.0hr, the floor slab 1.0hr.

The production class A, B factory buildings shall set up necessary pressure-relieving devices. The ratio of pressure-relieve area to class A building volume is 0.05, to class B is 0.03.

The weight which servers as pressure-relieve area should be not over 70kg/m².

2.3 The fire compartment area for warehouse building

Fire-resistance ratings, Maximum height and occupied area of warehouse building:

Class of stored goods	Maximum height(m)	Fire-resistance rating	Maximum floor area of fire division(m ²)		
			Single story	multistory	
				Two stories	Three stories and up
A	-	1, 2	5200	-	-
B	18	1, 2	7800	-	-
C	36	1, 2	10500	7800	5200
D	No limit	1, 2	No limit	10500	7800

The maximum admitted area of a fire division may be increased by one time, which an automatic sprinkler be installed.

The class A, B warehouse buildings having explosion hazard shall set up pressure-relieving devices.

The floor material of warehouse building should be non-combustible.

2.4 The fire compartment area for public building

Fire-resistance ratings, Maximum height and occupied area of public building:

Building fire-resistance class	Maximum stories	Maximum floor area of fire division (m ²)
1,2	6	2500

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The maximum admitted area of a fire division may be increased by one time, which an automatic sprinkler be installed.

2.5 Emergency evacuation

Sliding doors, roller shutter doors, revolving doors shall not be applied as evacuation doors.

Fire-escape doors shall be so located that they are never more than 35m away from any point on the floor.

The distance between the bottom of a staircase and the fire exit shall not exceed 35m.

Escape corridors shall not include dead-ends greater than 10m long.

Fire-escape doors of the building shall open to the direction of evacuation.

Fire-resistance ratings of metal stud plasterboard partition should be 1.25hr.

The wide of the escape corridor shall not be less than 1.4m.

The height of the exiting door on escape corridor to outside shall not be lower than 2.0m.

The number of safety exits shall not be less than two in follow case:

- In class A and B factory building, the floor area is more than 100m² or the number of workers is more than 5;
- In other class factory buildings, the floor area is more than 1000m² or the number of workers is more than 25;
- The number of exiting doors in public building shall not be less than two when the occupants init more than 50.

The corridor long than 60m shall be separated with the wall of rating 1.0hr, the door 0.5hr.

2.6 Evacuation staircase

Type of escape stair :

- Enclosed staircase;
- Non-enclosed staircase ;
- Outdoor evacuation stair ;
- Smoke-proof staircase;

The stair of production building shall adopt enclosed staircase.

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The wide of evacuation stair shall not be less than 1.2m.

The outdoor evacuation stair shall conform to the following requirements:

- The material of the stair shall be non-combustible.
- The fire rating of the wall shall not be less than 2.0hr.
- The handrail of the outdoor stair shall not be less than 1.2m high. No windows shall be set up on the wall within 1m around the stair.

2.7 Others

Cavity fire barriers shall be installed where vertical fire separation between rooms is required by either the Client requirements or local Regulation. The barriers shall provide complete fire and smoke separation between the areas. Cavity barriers, trims and accessories shall be compatible and of a type recommended by the suspended ceiling manufacturer.

Raised floors (computer floors) shall have 0.5 hr.